Hardware-Aware Quantum Bayesian Learner Compression

Linking Human Belief Updating to NISQ Circuit Complexity

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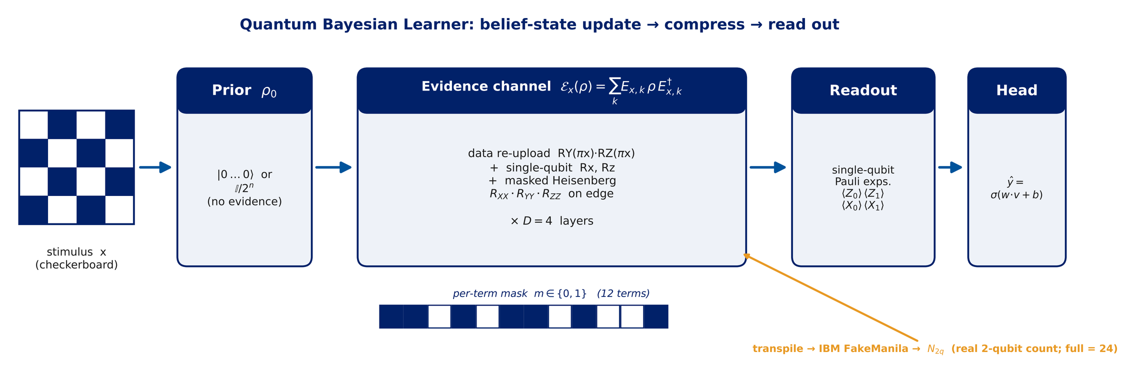
[1] Motivation — Two fields, one question

- Bayesian models cast human learning as **belief updating**: a prior over hypotheses, revised by evidence. **Quantum cognition** represents beliefs as **density operators** in Hilbert space, with evidence acting through **quantum channels**.
- On **NISQ** hardware, **two-qubit gates dominate the error budget**, so removing them without breaking the task is a central engineering goal.

Central question. *How much of a quantum Bayesian learner's entangling structure can we remove before it can no longer represent the task — and what happens on real hardware?*

[2] The Quantum Bayesian Learner



Left→right pipeline: stimulus x → prior ρ_0 → masked Heisenberg evidence channel (×D=4) → single-qubit Pauli readout → trainable head. The 12-cell mask gates the entanglers; transpiling to IBM FakeManila yields the real two-qubit cost N_{2q}.

- Belief = state ρ on a **2-qubit** register; prior ρ_0 = "no evidence yet." A stimulus x updates belief via a **CPTP evidence channel** $\mathcal{E}_x(\rho)=\sum_k E_{xk}\,\rho\,E_{xk}^\dagger$.
- Update circuit = **hardware-efficient ansatz**: data re-uploading + per-layer single-qubit R_x, R_z + **Heisenberg entanglers** $U_{ij}=\exp[-i(\alpha X_iX_j+\beta Y_iY_j+\gamma Z_iZ_j)]$ as native R_{XX}, R_{YY}, R_{ZZ}. Depth $D=4$, 2 re-uploads.
- **Readout: single-qubit Pauli expectations** $v=(\langle Z_0\rangle,\langle Z_1\rangle,\langle X_0\rangle,\langle X_1\rangle)\to$ trainable head $\hat{y}=\sigma(w\cdot v+b)$. Single-qubit-only readout means **entanglement is the ONLY way to carry feature interaction** — the two-qubit budget literally bounds the learner's capacity.
- Two realizations, same conclusions: a fast **pure-state** model and a literal **mixed-state density matrix** (maximally mixed prior $I/2^n$, non-unital Lüders/POVM evidence filter + amplitude damping, Bayesian trace renormalization).

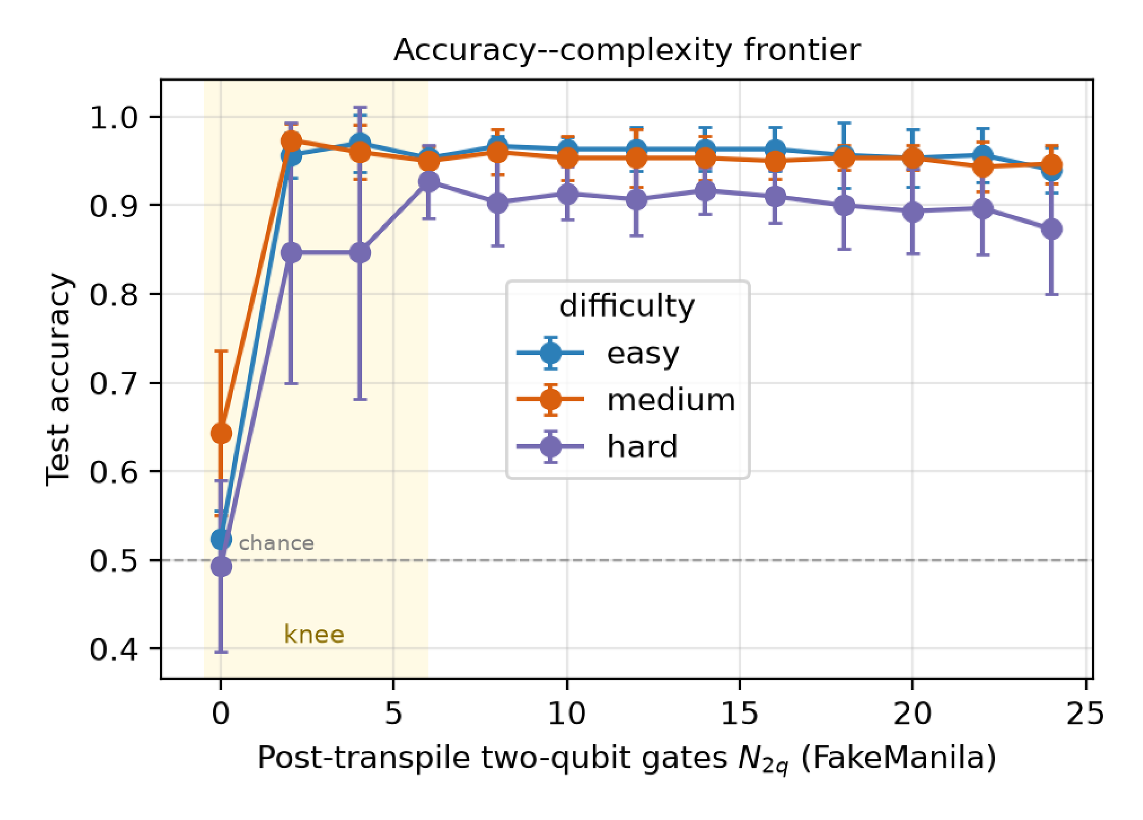
[3] Method — mask, real cost, greedy compression

- **Per-term binary mask** $m_{\ell,p}\in\{0,1\}$ over $p\in\{XX,YY,ZZ\}$ in each of D=4 layers ⇒ **12 maskable terms** ($R_{PP}(0)=I$ when off).
- **Hardware-aware cost** N_{2q} = real post-transpile two-qubit gate count on IBM **FakeManila** (5-qubit, includes SWAPs), from Qiskit's transpiler. Full circuit ⇒ $N_{2q}=24$ (each active Heisenberg term = 2 CNOTs).
- **Objective:** $L(\theta,m)=\frac{1}{N}\sum_k \mathrm{BCE}(y_k,\hat{y})+\lambda\,N_{2q}(m)$.
- **Greedy structured pruning:** from the full circuit, repeatedly drop the term whose removal least increases training cross-entropy, **warm-start retrain** — tracing the accuracy-vs-N_{2q} **frontier** from 24 down to 0.
- Exact **JAX autodiff** (no parameter-shift / finite differences); statevector verified **bit-for-bit vs Qiskit** ($|1-\mathrm{fidelity}|\approx 4\times10^{-16}$).

Task — controllable difficulty knob.

- **Checkerboard** categorization: $y=(\lfloor fx_0\rfloor+\lfloor fx_1\rfloor)\bmod 2$, stimuli in $[0,1]^2$.
- Difficulty f: f=2 **easy** / f=3 **medium** / f=4 **hard**. 200 stimuli, 70/30 split, balanced ⇒ **chance** = 0.5. The rule depends on the **joint** features — needing interaction only entanglement can supply.

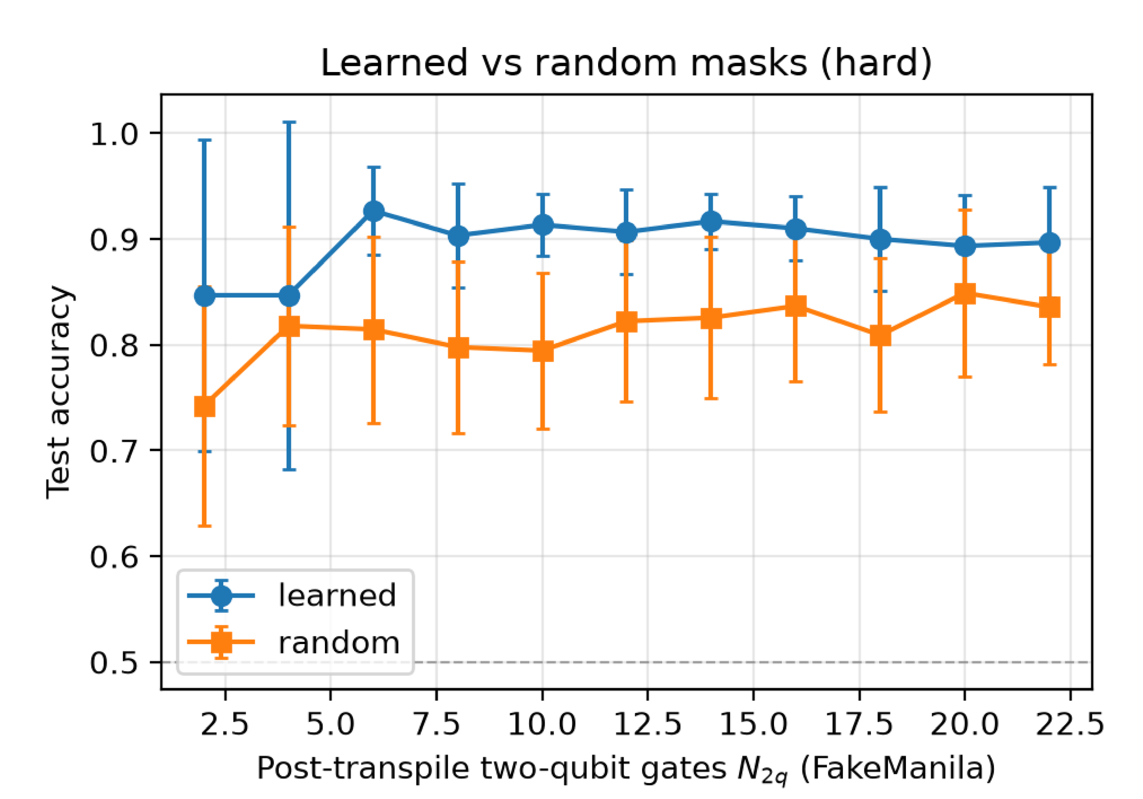
[4] Result 1 — The capacity boundary



Accuracy–complexity frontier across difficulties. Flat above the knee; collapse toward chance at $N_{2q}=0$.

- Above chance at all nonzero budgets (easy/medium ≈ **0.95–0.97**, hard ≈ **0.87–0.93**); frontier **nearly flat** — most entangling structure is **redundant**.
- At $N_{2q}=0$ all difficulties **collapse toward chance** (0.52 easy, 0.64 medium, 0.49 hard) — some entanglement is **necessary**, but far less than the full circuit.
- Hard task (not saturated): accuracy **peaks at intermediate cost** (**0.927** @ N_{2q}=6) and is lower for the full circuit (**0.873** @ N_{2q}=24) — moderate compression acts as a capacity-control **regularizer**.

[5] Result 2 — Structure beats count



Learned (greedy) masks vs random masks at matched N_{2q}, hard task.

- Learned masks vs **random masks at matched** N_{2q}: learned wins at **every** budget by ∼**6–12 points**.
- Per seed: learned beats random in **49 / 55** paired (seed, budget) comparisons, mean advantage +0.083, one-sided Wilcoxon $p<10^{-7}$.
- ⇒ **Which** entanglers you keep matters, not just how many.

N_{2q}	Learne	Randor	Δ
2	0.847	0.742	**+0.104**
6	**0.927**	0.814	**+0.112**
10	0.913	0.794	**+0.119**
14	0.917	0.826	**+0.091**
18	0.900	0.809	**+0.091**
22	0.897	0.836	**+0.061**

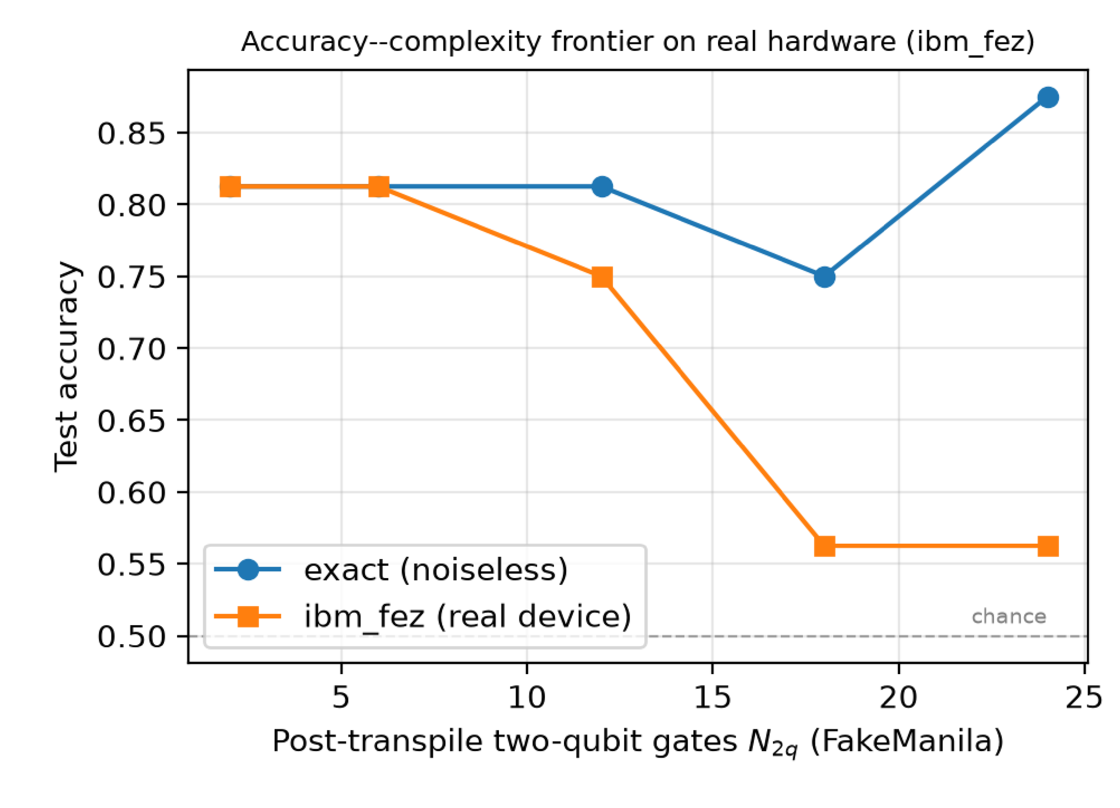
Hard task — learned vs random, mean accuracy, 5 seeds.

λ-sweep (hard): λ=0 → N_{2q} 9.6, acc 0.923; λ=0.04 → N_{2q} 3.2, acc 0.933; λ=0.15 → N_{2q} 1.6, acc 0.823. *Sheds ∼2/3 of 2-qubit gates for free, >90% at modest cost.*

[6] Robustness — not an artifact

- **Device noise model** (FakeManila, 4096 shots, 5 seeds): noisy accuracy tracks the noiseless frontier within a few % at every budget — savings genuine net of noise.
- **Density-matrix model** (genuine mixed state, non-unital channel): reproduces every finding — flat frontier, collapse at $N_{2q}=0$, hard peak **0.88 @ 6 vs 0.87**

[7] Result 3 — On REAL hardware, compression becomes NECESSARY



Centerpiece. Trained circuits on physical IBM `ibm_fez` *(156-qubit Heron). In simulation the frontier is flat; on hardware it falls monotonically.*

- Hard task, 5 device budgets, **16 test stimuli @ 2048 shots** (endpoints: 40 stimuli @ 4096 shots).
- On hardware accuracy **falls monotonically** with N_{2q}: **0.81** @ N_{2q}=2, 6 → **0.75 @ 12** → **collapses to 0.56** (near chance) **@ 18,24**.
- Endpoint confirmation: **full circuit 0.60** (sim 0.95) vs **compressed 0.83** (sim 0.93).

Takeaway: On real NISQ hardware, hardware-aware compression of the quantum Bayesian learner is not merely economical but NECESSARY — surplus entangling capacity is actively harmful, and task-aware pruning is what makes the learner usable at all.

[8] Key takeaways

- Most entanglement is **redundant** — sheds ∼**2/3** of two-qubit gates for **free**, >**90%** at modest cost.
- Entanglement is **necessary** — removing all of it collapses the learner to chance: a concrete **capacity boundary**.
- **Structure beats count** — learned masks beat random at every matched budget.
- **On real hardware, compression is what makes the learner work at all.**

Future work. Multi-qubit belief registers & higher-dim stimuli; richer non-unital evidence channels; hardware-aware objectives weighting depth, coherence & native fidelity; feeding measured device error rates back into the compression objective.

References & links

Key references

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Repository

github.com/zoraizmohammad/qb-learner-compression

Acknowledgments

Device runs used IBM Quantum services. The author thanks Duke University's Pratt School of Engineering.

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